TASK 5: PAYMENT TRANSACTION MANAGEMENT SYSTEM

PROJECT DETAILS

- **Name:** Kaviya S
- **Roll No:** E25AI118
- **Department:** B.Sc Computer Science With AI

SCENARIO

The e-commerce bookstore (**PALETTE: Pixel & Pages**) needs to maintain payment information for customer orders. The system records payment details, tracks transaction status, manages successful and failed payments, and generates comprehensive payment-related business reports.

1. CREATE PAYMENT TABLE

i. Payment Entity Structure:

Attribute	Description	Data Type	Constraints
Payment_ID	Unique payment identifier	VARCHAR2(7)	Primary Key
Order_ID	Order reference	VARCHAR2(10)	Foreign Key (Orders)
Payment_Date	Date of transaction	DATE	NOT NULL
Payment_Mode	UPI / Credit Card / Debit Card / Net Banking / COD	VARCHAR2(15)	NOT NULL
Payment_Status	Completed / Failed / Pending	VARCHAR2(10)	CHECK ('Completed', 'Failed', 'Pending')
Transaction_Amount	Paid amount	NUMBER(10,2)	CHECK (> 0)

ii. Payment Table Creation Query:

```
SQL> CREATE TABLE Payment (
  2      Payment_ID VARCHAR2(7),
  3      Order_ID VARCHAR2(10),
  4      Payment_Date DATE NOT NULL,
  5      Payment_Mode VARCHAR2(15) NOT NULL,
  6      Payment_Status VARCHAR2(10)
  7          CHECK (Payment_Status IN ('Completed', 'Failed', 'Pending')),
  8      Transaction_Amount NUMBER(10,2)
  9          CHECK (Transaction_Amount > 0),
 10      PRIMARY KEY (Payment_ID),
 11      FOREIGN KEY (Order_ID) REFERENCES Orders (Order_ID)
 12  );

Table created.
```

2. DEFINE RELATIONSHIPS AND CONSTRAINTS

i. Relationships:

- Each Order can have payment transaction records linked to it ($1 \rightarrow *$).
- Order_ID is the Primary Key in the Orders table and acts as a Foreign Key in the Payment table, linking payments directly to customer orders.
- Payment_ID uniquely identifies each payment record.

ii. Constraint Explanations:

- **PRIMARY KEY:** Uniquely identifies each payment record (Payment_ID).
- **FOREIGN KEY:** Links the Payment table back to the parent Orders table (Order_ID).
- **NOT NULL / CHECK:** Guarantees transaction validity, appropriate status states ('Completed', 'Failed', 'Pending'), and positive transaction amounts (> 0).

3. STORE PAYMENT DETAILS

i. Insert Statements:

```
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY001','ORD-501',TO_DATE('01-MAR-26','DD-MON-RR'),'Credit Card','Completed',1098);

1 row created.

SQL>
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY002','ORD-502',TO_DATE('05-MAR-26','DD-MON-RR'),'UPI','Completed',1250);

1 row created.

SQL>
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY003','ORD-503',TO_DATE('10-MAR-26','DD-MON-RR'),'Debit Card','Pending',599);

1 row created.

SQL>
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY004','ORD-504',TO_DATE('12-MAR-26','DD-MON-RR'),'Net Banking','Completed',1149);

1 row created.

SQL>
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY005','ORD-505',TO_DATE('15-MAR-26','DD-MON-RR'),'UPI','Completed',699);

1 row created.

SQL>
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY006','ORD-506',TO_DATE('16-MAR-26','DD-MON-RR'),'Credit Card','Pending',799);

1 row created.

SQL>
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY007','ORD-507',TO_DATE('17-MAR-26','DD-MON-RR'),'UPI','Completed',499);

1 row created.

SQL>
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY008','ORD-508',TO_DATE('18-MAR-26','DD-MON-RR'),'Debit Card','Completed',599);

1 row created.

SQL>
SQL> INSERT INTO PAYMENT VALUES
  2  ('PAY009','ORD-509',TO_DATE('19-MAR-26','DD-MON-RR'),'Net Banking','Completed',450);

1 row created.
```

ii. Display Payment Table:

```
SQL> SELECT * FROM PAYMENT;

PAYMENT ORDER_ID   PAYMENT_D PAYMENT_MODE    PAYMENT_ST TRANSACTION_AMOUNT
------- ---------- --------- --------------- ---------- ------------------
PAY001  ORD-501    01-MAR-26 Credit Card     Completed                1098
PAY002  ORD-502    05-MAR-26 UPI             Completed                1250
PAY003  ORD-503    10-MAR-26 Debit Card      Pending                   599
PAY004  ORD-504    12-MAR-26 Net Banking     Completed                1149
PAY005  ORD-505    15-MAR-26 UPI             Completed                 699
PAY006  ORD-506    16-MAR-26 Credit Card     Pending                   799
PAY007  ORD-507    17-MAR-26 UPI             Completed                 499
PAY008  ORD-508    18-MAR-26 Debit Card      Completed                 599
PAY009  ORD-509    19-MAR-26 Net Banking     Completed                 450
PAY010  ORD-510    20-MAR-26 UPI             Pending                   650
PAY011  ORD-511    21-MAR-26 Credit Card     Completed                 999

PAYMENT ORDER_ID   PAYMENT_D PAYMENT_MODE    PAYMENT_ST TRANSACTION_AMOUNT
------- ---------- --------- --------------- ---------- ------------------
PAY012  ORD-512    22-MAR-26 Debit Card      Completed                 850
PAY013  ORD-513    23-MAR-26 UPI             Pending                   550
PAY014  ORD-514    24-MAR-26 Credit Card     Completed                 799
PAY015  ORD-515    25-MAR-26 Net Banking     Completed                 899
PAY016  ORD-516    26-MAR-26 UPI             Failed                    450
PAY017  ORD-517    27-MAR-26 Credit Card     Completed                 699
PAY018  ORD-518    28-MAR-26 Debit Card      Completed                 750
PAY019  ORD-519    29-MAR-26 UPI             Pending                   399
PAY020  ORD-520    30-MAR-26 Net Banking     Completed                 550

20 rows selected.
```

4. MANAGE SUCCESSFUL AND FAILED TRANSACTIONS

i. Display all successful payments:

```
SQL> SELECT *
  2  FROM Payment
  3  WHERE Payment_Status = 'Completed';

PAYMENT ORDER_ID   PAYMENT_D PAYMENT_MODE    PAYMENT_ST TRANSACTION_AMOUNT
------- ---------- --------- --------------- ---------- ------------------
PAY001  ORD-501    01-MAR-26 Credit Card     Completed                1098
PAY002  ORD-502    05-MAR-26 UPI             Completed                1250
PAY004  ORD-504    12-MAR-26 Net Banking     Completed                1149
PAY005  ORD-505    15-MAR-26 UPI             Completed                 699
PAY007  ORD-507    17-MAR-26 UPI             Completed                 499
PAY008  ORD-508    18-MAR-26 Debit Card      Completed                 599
PAY009  ORD-509    19-MAR-26 Net Banking     Completed                 450
PAY011  ORD-511    21-MAR-26 Credit Card     Completed                 999
PAY012  ORD-512    22-MAR-26 Debit Card      Completed                 850
PAY014  ORD-514    24-MAR-26 Credit Card     Completed                 799
PAY015  ORD-515    25-MAR-26 Net Banking     Completed                 899

PAYMENT ORDER_ID   PAYMENT_D PAYMENT_MODE    PAYMENT_ST TRANSACTION_AMOUNT
------- ---------- --------- --------------- ---------- ------------------
PAY017  ORD-517    27-MAR-26 Credit Card     Completed                 699
PAY018  ORD-518    28-MAR-26 Debit Card      Completed                 750
PAY020  ORD-520    30-MAR-26 Net Banking     Completed                 550

14 rows selected.
```

ii. Find failed transactions:

```
SQL> SELECT * FROM Payment WHERE Payment_Status = 'Failed';

PAYMENT ORDER_ID   PAYMENT_D PAYMENT_MODE    PAYMENT_ST TRANSACTION_AMOUNT
------- ---------- --------- --------------- ---------- ------------------
PAY016  ORD-516    26-MAR-26 UPI             Failed                    450
```

iii. Count total successful and failed payments:

```
SQL> SELECT Payment_Status, COUNT(Payment_Status) FROM Payment GROUP BY Payment_Status;

PAYMENT_ST COUNT(PAYMENT_STATUS)
---------- ---------------------
Failed                         1
Completed                     14
Pending                        5
```

5. ANALYZE PAYMENT METHODS USED BY CUSTOMERS

Report 1: Payment Mode Analysis

```
SQL> SELECT COUNT(*) AS UPI_TRANSACTIONS FROM Payment WHERE Payment_Mode = 'UPI';

UPI_TRANSACTIONS
----------------
               7
```

```
SQL> SELECT COUNT(*) AS CARD_PAYMENTS FROM Payment WHERE Payment_Mode IN ('Credit Card', 'Debit Card');

CARD_PAYMENTS
-------------
            9
```

```
SQL> SELECT Payment_Mode, COUNT(*) AS TRANSACTION_COUNT FROM Payment GROUP BY Payment_Mode ORDER BY TRANSACTION_COUNT DESC;

PAYMENT_MODE    TRANSACTION_COUNT
--------------- -----------------
UPI                             7
Credit Card                     5
Debit Card                      4
Net Banking                     4
```

```
SQL> SELECT *
  2  FROM (
  3      SELECT Payment_Mode,
  4             COUNT(*) AS TRANSACTION_COUNT
  5      FROM Payment
  6      GROUP BY Payment_Mode
  7      ORDER BY COUNT(*) DESC
  8  )
  9  WHERE ROWNUM = 1;

PAYMENT_MODE    TRANSACTION_COUNT
--------------- -----------------
UPI                             7
```

Report 2: Revenue Analysis

```
SQL> SELECT SUM(Transaction_Amount) AS TOTAL_REVENUE FROM Payment WHERE Payment_Status = 'Completed';

TOTAL_REVENUE
-------------
        11290
```

```
SQL> SELECT Payment_Mode, SUM(Transaction_Amount) AS TOTAL_REVENUE FROM Payment WHERE Payment_Status = 'Completed' GROUP BY Payment_Mode ORDER BY TOTAL_REVENUE DESC;

PAYMENT_MODE    TOTAL_REVENUE
--------------- -------------
Credit Card              3595
Net Banking              3048
UPI                      2448
Debit Card               2199
```

```
SQL> SELECT ROUND(AVG(Transaction_Amount), 2) AS AVERAGE_TRANSACTION_AMOUNT FROM Payment WHERE Payment_Status = 'Completed';

AVERAGE_TRANSACTION_AMOUNT
--------------------------
                    806.43
```

Report 3: Customer Payment History

```
SQL> SELECT
  2      c.first_name || ' ' || c.last_name AS Customer_Name,
  3      o.Order_ID,
  4      o.Order_Date,
  5      p.Payment_Date,
  6      p.Payment_Mode,
  7      p.Transaction_Amount AS Amount,
  8      p.Payment_Status AS Payment_Status
  9  FROM customer c
 10  JOIN Orders o ON c.customer_id = o.Customer_ID
 11  JOIN Payment p ON o.Order_ID = p.Order_ID
 12  ORDER BY p.Payment_Date;

CUSTOMER_NAME                             ORDER_ID   ORDER_DAT PAYMENT_D
----------------------------------------- ---------- --------- ---------
PAYMENT_MODE        AMOUNT PAYMENT_ST
--------------- ---------- ----------
Ananya Roy                                ORD-501    01-MAR-26 01-MAR-26
Credit Card           1098 Completed

Ananya Roy                                ORD-502    05-MAR-26 05-MAR-26
UPI                   1250 Completed

Rohan Mehta                               ORD-503    10-MAR-26 10-MAR-26
Debit Card             599 Pending


CUSTOMER_NAME                             ORDER_ID   ORDER_DAT PAYMENT_D
----------------------------------------- ---------- --------- ---------
PAYMENT_MODE        AMOUNT PAYMENT_ST
--------------- ---------- ----------
Priya Nair                                ORD-504    12-MAR-26 12-MAR-26
Net Banking           1149 Completed

Aarav Sharma                              ORD-505    15-MAR-26 15-MAR-26
UPI                    699 Completed

Ishita Menon                              ORD-506    16-MAR-26 16-MAR-26
Credit Card            799 Pending


CUSTOMER_NAME                             ORDER_ID   ORDER_DAT PAYMENT_D
----------------------------------------- ---------- --------- ---------
PAYMENT_MODE        AMOUNT PAYMENT_ST
--------------- ---------- ----------
Kabir Singh                               ORD-507    17-MAR-26 17-MAR-26
UPI                    499 Completed

Diya Patel                                ORD-508    18-MAR-26 18-MAR-26
Debit Card             599 Completed

Arjun Nair                                ORD-509    19-MAR-26 19-MAR-26
Net Banking            450 Completed
```

```
CUSTOMER_NAME                          ORDER_ID   ORDER_DAT PAYMENT_D
-------------------------------------- ---------- --------- ---------
PAYMENT_MODE        AMOUNT PAYMENT_ST
--------------- ---------- ----------
Meera Rao                              ORD-510    20-MAR-26 20-MAR-26
UPI                    650 Pending

Rohan Verma                            ORD-511    21-MAR-26 21-MAR-26
Credit Card            999 Completed

Anika Iyer                             ORD-512    22-MAR-26 22-MAR-26
Debit Card             850 Completed


CUSTOMER_NAME                          ORDER_ID   ORDER_DAT PAYMENT_D
-------------------------------------- ---------- --------- ---------
PAYMENT_MODE        AMOUNT PAYMENT_ST
--------------- ---------- ----------
Nikhil Bose                            ORD-513    23-MAR-26 23-MAR-26
UPI                    550 Pending

Pooja Reddy                            ORD-514    24-MAR-26 24-MAR-26
Credit Card            799 Completed

Vikram Kapoor                          ORD-515    25-MAR-26 25-MAR-26
Net Banking            899 Completed


CUSTOMER_NAME                          ORDER_ID   ORDER_DAT PAYMENT_D
-------------------------------------- ---------- --------- ---------
PAYMENT_MODE        AMOUNT PAYMENT_ST
--------------- ---------- ----------
Sneha Gupta                            ORD-516    26-MAR-26 26-MAR-26
UPI                    450 Failed

Aditya Mishra                          ORD-517    27-MAR-26 27-MAR-26
Credit Card            699 Completed

Kavya Das                              ORD-518    28-MAR-26 28-MAR-26
Debit Card             750 Completed


CUSTOMER_NAME                          ORDER_ID   ORDER_DAT PAYMENT_D
-------------------------------------- ---------- --------- ---------
PAYMENT_MODE        AMOUNT PAYMENT_ST
--------------- ---------- ----------
Rahul Joseph                           ORD-519    29-MAR-26 29-MAR-26
UPI                    399 Pending

Neha Varma                             ORD-520    30-MAR-26 30-MAR-26
Net Banking            550 Completed
```

Payment Analysis Matrix:

```
SQL> SELECT
  2      Payment_Mode,
  3      COUNT(*) AS TRANSACTION_COUNT,
  4      SUM(Transaction_Amount) AS TOTAL_REVENUE,
  5      ROUND(AVG(Transaction_Amount), 2) AS AVG_TRANSACTION
  6  FROM Payment
  7  WHERE Payment_Status = 'Completed'
  8  GROUP BY Payment_Mode
  9  ORDER BY TOTAL_REVENUE DESC;

PAYMENT_MODE    TRANSACTION_COUNT TOTAL_REVENUE AVG_TRANSACTION
--------------- ----------------- ------------- ---------------
Credit Card                     4          3595          898.75
Net Banking                     4          3048             762
UPI                             3          2448             816
Debit Card                      3          2199             733
```